

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Deep learning [NPT_4124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare loss functions used in classification and regression tasks and explain how they influence optimization behavior. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Evaluate how model interpretability differs between regression and classification problems in real-world decision-making scenarios. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2) Compare the impact of shallow and deep architectures in feedforward networks. What advantages does increasing the depth of the network provide? (5)

A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Analyze the relationship between the gradient of the loss function and the optimization of the weights in a neural network during backpropagation. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3) Evaluate how augmentation strategies differ for image, text, and audio data, and their impact on model robustness. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) Evaluate the role of dropout as a regularization technique in neural networks. How does it impact the training process and model generalization? (5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 4) Evaluate how CNN architectures like ResNet or Inception overcome vanishing gradient and computational efficiency challenges. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Examine how filter size, stride, and padding affect the receptive field and feature extraction in CNNs. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5) Explain LSTMs manage cell states and gates to retain information over long sequences. (5)

A) Course Outcome: "CO2" / Cognitive Level: "Analysis"

B) Evaluate LSTM performance in real-world applications such as speech recognition or sentiment analysis compared to standard RNNs. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4124:Deep learning

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	15
3	4	CO 4	20
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4124:Deep learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	1	5(10%)
4	Analysis	7	35(70%)
5	Create	2	10(20%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)